

Richness Algae Metacommunity

Eros Fernando Geppi

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Analisis de Biodiversidad de Metacomunidades de algas en Canarias 1992-2021

```
# Visualización elegante y flexible
library(ggplot2)      # Gráficos estáticos usando gramática de gráficos

# Manipulación de datos
library(dplyr)        # Manipulación de datos (filtrar, seleccionar, resumir, etc.)
library(tidyr)        # Transformación de datos de formato ancho/largo
library(knitr)        # Transformación de datos de formato ancho/largo
library(forcats)      # Manipulación de factores (categorías)
library(purrr)        # Programación funcional (aplicación de funciones a listas)
library(broom)        # Convierte resultados estadísticos en data frames ordenados

# Lectura y organización de archivos
library(readxl)       # Importación de archivos Excel (.xls, .xlsx)
library(here)         # Gestión robusta de rutas relativas a la raíz del proyecto

# Temas y colores para gráficos
library(hrbrthemes)   # Temas modernos para ggplot2
library(viridis)      # Paletas de colores perceptualmente uniformes

# Análisis estadístico y ecológico
library(vegan)        # Análisis de ecología multivariante (diversidad, NMDS, etc.)
library(cluster)      # Métodos de agrupamiento (clustering)
library(gam)          # Modelos aditivos generalizados (GAM) - versión antigua
library(mgcv)         # Modelos aditivos generalizados (GAM) más actualizada y robusta
library(tweedie)      # Modelado de distribuciones de la familia Tweedie

# Visualización avanzada y combinaciones de gráficos
library(patchwork)    # Combina múltiples gráficos de ggplot2 en un solo panel

# Exportación de gráficos y reportes
library(officer)      # Generación y edición de documentos Word/PowerPoint
library(rvg)          # Exporta gráficos como vectores editables a PowerPoint/Word
library(export)       # Exporta gráficos de R a varios formatos (PowerPoint, PNG, etc.)
library(tibble)
```

1. Preparacion inicial de los datos

```
##           Reino           Filo           Clase
## Chromista:21287 Charophyta : 21 Florideophyceae :29559
## Plantae :43103 Chlorophyta:13235 Phaeophyceae :21222
##           Ochrophyta :21287 Ulvophyceae :11022
##           Rhodophyta :29847 Chlorophyceae : 2053
##           Trebouxioephyceae : 157
##           Compsopogonophyceae: 117
##           (Other) : 260
##           Orden           Familia           Genero
## Ceramiales :14143 Dictyotaceae : 7628 Gongolaria: 4017
## Fucales : 8879 Sargassaceae : 7501 Caulerpa : 3908
## Dictyotales : 7628 Rhodomelaceae: 7405 Dictyota : 2498
## Bryopsidales : 5838 Caulerpaceae : 3908 Cystoseira: 2483
## Corallinales : 3706 Corallinaceae: 2697 Gelidium : 2302
## Cladophorales: 3286 Gelidiaceae : 2302 Dunaliella: 1704
## (Other) :20910 (Other) :32949 (Other) :47478
##           Specie           Year           Isla
## Gongolaria abiesmarina: 3874 2005 :29986 El Hierro : 8343
## Dunaliella salina : 1704 2008 : 7319 Fuerteventura: 7515
## Gelidium canariense : 1515 2014 : 6021 Gran Canaria : 8794
## Lobophora variegata : 1481 2011 : 2797 La Gomera : 1594
## Caulerpa prolifera : 1475 1996 : 2602 La Palma : 8566
## Alsidium corallinum : 1341 2001 : 2221 Lanzarote :20756
## (Other) :53000 (Other):13444 Tenerife : 8822
##           Lat           Lon
## Min. :33.71 Min. : -18.41
## 1st Qu.:34.20 1st Qu.: -17.94
## Median :34.78 Median : -15.54
## Mean :34.70 Mean : -15.59
## 3rd Qu.:35.18 3rd Qu.: -13.67
## Max. :35.72 Max. : -13.17
##
```

1.1. Revision del numero de niveles por categoría

```
Data %>%
  summarise(
    n_Realm = n_distinct(Reino),
    n_File = n_distinct(Filo),
    n_Class = n_distinct(Clase),
    n_Order = n_distinct(Orden),
    n_Family = n_distinct(Familia),
    n_Gender = n_distinct(Genero),
    n_Species = n_distinct(Specie),
    n_Islands = n_distinct(Isla),
    n_Years = n_distinct(Year)
  ) %>% kable()
```

n_Realm	n_File	n_Class	n_Order	n_Family	n_Gender	n_Species	n_Islands	n_Years
2	4	14	49	111	289	573	7	30

1.2. Tabla de frecuencias: Registros por año.

```
Data %>% count(Year) %>%
  arrange(desc(n)) %>%
  print(n=Inf) %>%
  kable()
```

```
## # A tibble: 30 x 2
##   Year      n
##   <fct> <int>
## 1 2005 29986
## 2 2008  7319
## 3 2014  6021
## 4 2011  2797
## 5 1996  2602
## 6 2001  2221
## 7 1993  1615
## 8 2003  1589
## 9 2002  1504
## 10 1998  1396
## 11 2000   846
## 12 1999   749
## 13 1992   714
## 14 2004   707
## 15 2015   581
## 16 1994   463
## 17 2021   442
## 18 2007   426
## 19 2018   418
## 20 2012   343
## 21 1997   317
## 22 2009   306
## 23 2006   290
## 24 1995   248
## 25 2013   163
## 26 2017   108
## 27 2016    62
## 28 2010    59
## 29 2019    55
## 30 2020    43
```

Year	n
2005	29986
2008	7319
2014	6021
2011	2797

Year	n
1996	2602
2001	2221
1993	1615
2003	1589
2002	1504
1998	1396
2000	846
1999	749
1992	714
2004	707
2015	581
1994	463
2021	442
2007	426
2018	418
2012	343
1997	317
2009	306
2006	290
1995	248
2013	163
2017	108
2016	62
2010	59
2019	55
2020	43

1.3. Tabla de frecuencias: Registros por especie

```
Data %>% count(Specie) %>%
  arrange(desc(n)) %>%
  head(15) %>%
  kable()
```

Specie	n
Gongolaria abiesmarina	3874
Dunaliella salina	1704
Gelidium canariense	1515
Lobophora variegata	1481
Caulerpa prolifera	1475
Alsidium corallinum	1341
Padina pavonica	1334
Fucus spiralis	1332
Caulerpa racemosa	1258
Lophocladia trichoclados	1168
Halopteris scoparia	1112
Caulerpa webbiana	1088
Jania pedunculata	1061
Corallina ferreyrae	1056

Specie	n
Cystoseira compressa	1014

1.4. Detección de NA o registros sin ubicación

```
Data %>%
  summarise(
    na_lat = sum(is.na(Lat)),
    na_lon = sum(is.na(Lon)),
    na_year = sum(is.na(Year)),
    na_isla = sum(is.na(Isla)),
    na_specie = sum(is.na(Specie))
  ) %>%
  kable()
```

na_lat	na_lon	na_year	na_isla	na_specie
0	0	0	0	0

2. Evaluacion de diversidad alpha, beta y patrones espaciotemporales.

2.1. Diversidad Alpha por sitio y año.

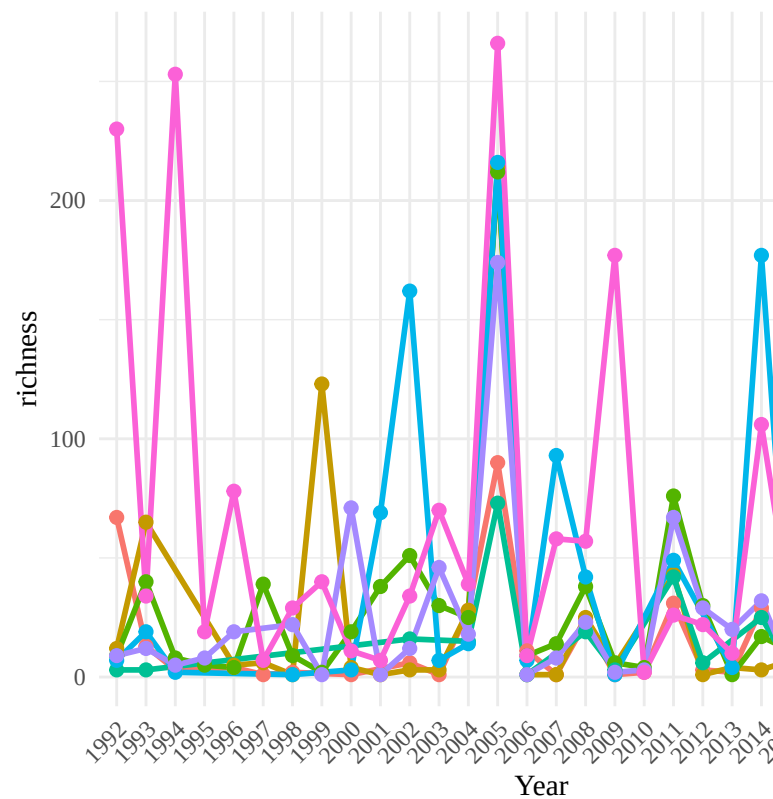
La *riqueza alfa* es el número de especies diferentes en una unidad local.

```
alpha_div <- Data %>%
  group_by(Isla, Year) %>%
  summarise(richness = n_distinct(Specie)) %>%
  ungroup()
```

2.1.1. Calculo de diversidad alpha por isla y año

```
ggplot(alpha_div, aes(x= Year, y= richness, group = Isla, color=Isla)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  theme_minimal() +
  labs(title = "Alpha richness (nº of species) by year and island") +
  theme(axis.text.x = element_text( angle = 45, hjust=+1),
        text = element_text(family = "serif")
  )
```

Alpha richness (nº of species) by year and island



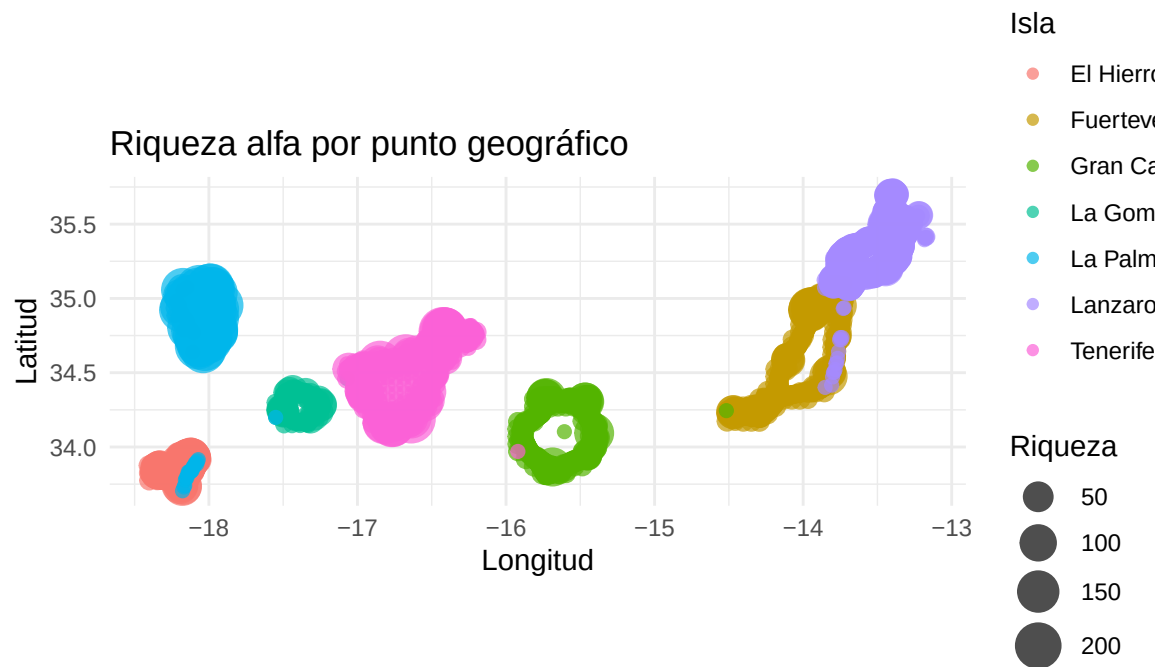
2.1.2 Visualizacion de riqueza local por isla y año.

2.2. Mapa de riqueza por sition

```
alpha_geo <- Data %>%
  group_by(Isla, Lat, Lon) %>%
  summarise(richness = n_distinct(Specie)) %>%
  ungroup()
```

2.2.1. Agregar diversidad alpha por coordenadas

```
ggplot(alpha_geo, aes(x = Lon, y = Lat)) +
  geom_point(aes(size = richness, color = Isla), alpha = 0.7) +
  scale_size_continuous(range = c(2, 8)) +
  theme_minimal() +
  coord_fixed() +
  labs(title = "Riqueza alfa por punto geográfico",
       x = "Longitud", y = "Latitud", size = "Riqueza")
```



2.2.2. Visualizacion

2.3. Diversidad beta entre islas

La *diversidad beta* mide el recambio de especies entre sitios. Para esto generaremos una matriz presencia-ausencia por isla.

```
beta_matrix <- Data %>%
  group_by(Isla, Specie) %>%
  summarise(present = 1, .groups = "drop") %>%
  pivot_wider(names_from = Specie, values_from = present, values_fill = 0)

# Extraer solo la matriz de especies
beta_mat <- as.data.frame(beta_matrix[, -1])
rownames(beta_mat) <- beta_matrix$Isla
```

2.3.1. Matriz de Comunidad

```
jaccard_dist <- vegdist(beta_mat, method = "jaccard")
```

2.3.2. Calcular distancias y disimilitudes (Jaccard)

```
nmds <- metaMDS(beta_mat, distance = "jaccard", k = 2, trymax = 100)
```

2.3.3. Análisis de ordenación (NMDS)

```
## Run 0 stress 0
## Run 1 stress 0.1373884
## Run 2 stress 9.511413e-05
## ... Procrustes: rmse 0.1100266 max resid 0.1480474
## Run 3 stress 8.871179e-05
## ... Procrustes: rmse 0.1951539 max resid 0.2861698
## Run 4 stress 9.863274e-05
## ... Procrustes: rmse 0.0631747 max resid 0.1138677
## Run 5 stress 9.362419e-05
## ... Procrustes: rmse 0.1146293 max resid 0.1681989
## Run 6 stress 0.1373884
## Run 7 stress 9.601077e-05
## ... Procrustes: rmse 0.06222063 max resid 0.1122324
## Run 8 stress 9.59273e-05
## ... Procrustes: rmse 0.1952658 max resid 0.2861217
## Run 9 stress 9.592856e-05
## ... Procrustes: rmse 0.1951466 max resid 0.2861664
## Run 10 stress 0.293306
## Run 11 stress 9.328474e-05
## ... Procrustes: rmse 0.1101781 max resid 0.1589272
## Run 12 stress 0.1373884
## Run 13 stress 9.880576e-05
## ... Procrustes: rmse 0.1951389 max resid 0.2861622
## Run 14 stress 8.296969e-05
## ... Procrustes: rmse 0.1952365 max resid 0.2861088
## Run 15 stress 9.486639e-05
## ... Procrustes: rmse 0.1094499 max resid 0.1589685
## Run 16 stress 0.2010294
## Run 17 stress 9.744373e-05
## ... Procrustes: rmse 0.1951402 max resid 0.286162
## Run 18 stress 0
## ... Procrustes: rmse 0.05240196 max resid 0.07480802
## Run 19 stress 9.572238e-05
## ... Procrustes: rmse 0.1427474 max resid 0.2087284
## Run 20 stress 9.345955e-05
## ... Procrustes: rmse 0.04341571 max resid 0.06612048
## Run 21 stress 0.2933061
## Run 22 stress 9.538327e-05
## ... Procrustes: rmse 0.1952626 max resid 0.2861106
## Run 23 stress 9.767577e-05
## ... Procrustes: rmse 0.195127 max resid 0.2860002
## Run 24 stress 9.62427e-05
## ... Procrustes: rmse 0.1940735 max resid 0.2843255
## Run 25 stress 9.52529e-05
## ... Procrustes: rmse 0.1015857 max resid 0.1394324
## Run 26 stress 0.293306
## Run 27 stress 9.994005e-05
## ... Procrustes: rmse 0.0475224 max resid 0.07222465
```



```

## Run 28 stress 9.876527e-05
## ... Procrustes: rmse 0.1951606 max resid 0.2861701
## Run 29 stress 9.559845e-05
## ... Procrustes: rmse 0.05037796 max resid 0.09134823
## Run 30 stress 9.995406e-05
## ... Procrustes: rmse 0.06315949 max resid 0.1138339
## Run 31 stress 0.2010294
## Run 32 stress 0.1373884
## Run 33 stress 0
## ... Procrustes: rmse 0.03749155 max resid 0.06855141
## Run 34 stress 0.1373884
## Run 35 stress 9.942542e-05
## ... Procrustes: rmse 0.1544848 max resid 0.2244156
## Run 36 stress 9.96888e-05
## ... Procrustes: rmse 0.1697103 max resid 0.2457857
## Run 37 stress 9.474759e-05
## ... Procrustes: rmse 0.1951463 max resid 0.2861654
## Run 38 stress 9.380262e-05
## ... Procrustes: rmse 0.1952582 max resid 0.2861078
## Run 39 stress 9.719709e-05
## ... Procrustes: rmse 0.1117932 max resid 0.1640354
## Run 40 stress 8.922008e-05
## ... Procrustes: rmse 0.07753843 max resid 0.1427527
## Run 41 stress 9.901505e-05
## ... Procrustes: rmse 0.1854022 max resid 0.2679115
## Run 42 stress 0
## ... Procrustes: rmse 0.02086795 max resid 0.03078965
## Run 43 stress 0.1373884
## Run 44 stress 9.526492e-05
## ... Procrustes: rmse 0.1951489 max resid 0.2861664
## Run 45 stress 8.750723e-05
## ... Procrustes: rmse 0.0757686 max resid 0.1040452
## Run 46 stress 9.3051e-05
## ... Procrustes: rmse 0.06315947 max resid 0.1138231
## Run 47 stress 9.251335e-05
## ... Procrustes: rmse 0.1519852 max resid 0.2208023
## Run 48 stress 5.482594e-05
## ... Procrustes: rmse 0.06774612 max resid 0.1040449
## Run 49 stress 9.070608e-05
## ... Procrustes: rmse 0.0698946 max resid 0.09619694
## Run 50 stress 9.393584e-05
## ... Procrustes: rmse 0.1951464 max resid 0.2861656
## Run 51 stress 9.511363e-05
## ... Procrustes: rmse 0.1952597 max resid 0.286105
## Run 52 stress 9.120032e-05
## ... Procrustes: rmse 0.1329033 max resid 0.1951266
## Run 53 stress 0.1373884
## Run 54 stress 9.287247e-05
## ... Procrustes: rmse 0.1952612 max resid 0.286111
## Run 55 stress 9.741242e-05
## ... Procrustes: rmse 0.1952599 max resid 0.2861047
## Run 56 stress 9.612045e-05
## ... Procrustes: rmse 0.04922015 max resid 0.07241797
## Run 57 stress 9.079672e-05

```

```

## ... Procrustes: rmse 0.1952599  max resid 0.2861553
## Run 58 stress 7.414832e-05
## ... Procrustes: rmse 0.1952373  max resid 0.2862122
## Run 59 stress 0.1373884
## Run 60 stress 8.342222e-05
## ... Procrustes: rmse 0.0667974  max resid 0.09535815
## Run 61 stress 0.1373884
## Run 62 stress 9.884872e-05
## ... Procrustes: rmse 0.1951541  max resid 0.2861689
## Run 63 stress 9.866611e-05
## ... Procrustes: rmse 0.1951502  max resid 0.2861695
## Run 64 stress 9.839172e-05
## ... Procrustes: rmse 0.1404654  max resid 0.190326
## Run 65 stress 8.818819e-05
## ... Procrustes: rmse 0.08526025  max resid 0.156076
## Run 66 stress 9.48134e-05
## ... Procrustes: rmse 0.063166  max resid 0.1138433
## Run 67 stress 9.182061e-05
## ... Procrustes: rmse 0.1952634  max resid 0.2861178
## Run 68 stress 9.906224e-05
## ... Procrustes: rmse 0.1652108  max resid 0.2396018
## Run 69 stress 9.258483e-05
## ... Procrustes: rmse 0.1952707  max resid 0.2861522
## Run 70 stress 9.128298e-05
## ... Procrustes: rmse 0.1033412  max resid 0.1498683
## Run 71 stress 9.861003e-05
## ... Procrustes: rmse 0.1952676  max resid 0.2861276
## Run 72 stress 9.560166e-05
## ... Procrustes: rmse 0.1212507  max resid 0.1740809
## Run 73 stress 9.796524e-05
## ... Procrustes: rmse 0.1952709  max resid 0.2861422
## Run 74 stress 9.409964e-05
## ... Procrustes: rmse 0.195109  max resid 0.2859816
## Run 75 stress 8.698281e-05
## ... Procrustes: rmse 0.1039205  max resid 0.147083
## Run 76 stress 9.896691e-05
## ... Procrustes: rmse 0.1951397  max resid 0.2861608
## Run 77 stress 9.513982e-05
## ... Procrustes: rmse 0.1952561  max resid 0.2861328
## Run 78 stress 8.481321e-05
## ... Procrustes: rmse 0.06274211  max resid 0.1119464
## Run 79 stress 0.1373884
## Run 80 stress 9.909891e-05
## ... Procrustes: rmse 0.06316081  max resid 0.1138343
## Run 81 stress 7.176932e-05
## ... Procrustes: rmse 0.1952409  max resid 0.2861992
## Run 82 stress 9.445835e-05
## ... Procrustes: rmse 0.1063489  max resid 0.1583821
## Run 83 stress 9.925256e-05
## ... Procrustes: rmse 0.1950667  max resid 0.2860042
## Run 84 stress 9.777433e-05
## ... Procrustes: rmse 0.1952623  max resid 0.2861055
## Run 85 stress 9.235928e-05
## ... Procrustes: rmse 0.195263  max resid 0.2861183

```

```

## Run 86 stress 0.2010294
## Run 87 stress 9.999527e-05
## ... Procrustes: rmse 0.1950658  max resid 0.286002
## Run 88 stress 9.756039e-05
## ... Procrustes: rmse 0.063171  max resid 0.1138586
## Run 89 stress 0.1785336
## Run 90 stress 0.1373884
## Run 91 stress 9.193192e-05
## ... Procrustes: rmse 0.1951491  max resid 0.2861659
## Run 92 stress 9.847657e-05
## ... Procrustes: rmse 0.1951397  max resid 0.2861625
## Run 93 stress 9.115493e-05
## ... Procrustes: rmse 0.1952625  max resid 0.2861165
## Run 94 stress 9.399102e-05
## ... Procrustes: rmse 0.1668645  max resid 0.2419307
## Run 95 stress 9.151756e-05
## ... Procrustes: rmse 0.1952628  max resid 0.2861138
## Run 96 stress 0.1785336
## Run 97 stress 9.562093e-05
## ... Procrustes: rmse 0.1951439  max resid 0.2861649
## Run 98 stress 6.108815e-05
## ... Procrustes: rmse 0.01923296  max resid 0.03263891
## Run 99 stress 0.1373884
## Run 100 stress 9.193973e-05
## ... Procrustes: rmse 0.1952524  max resid 0.2861347
## *** Best solution was not repeated -- monoMDS stopping criteria:
##      80: stress < smin
##      6: stress ratio > sratmax
##     14: scale factor of the gradient < sfgrmin

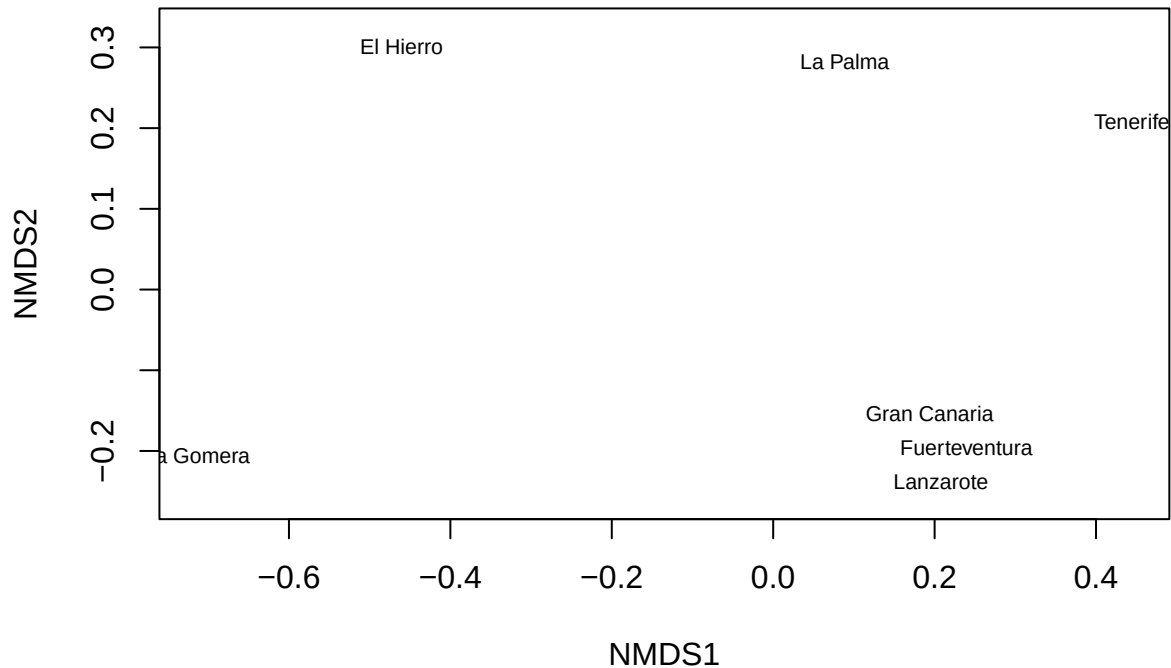
```

```

plot(nmds, type = "t", display = "sites", main = "NMDS - Diversidad beta entre islas")

```

NMDS – Diversidad beta entre islas



2.4 — Diversidad beta temporal dentro de islas.

La idea aquí es evaluar el *recambio de especies a lo largo del tiempo* en cada isla. Es el mismo enfoque de comunidad presencia/ausencia, pero ahora por combinación Isla + Año.

```
# Construimos la variable combinada Isla_Year
Data <- Data %>%
  mutate(Isla_Year = paste(Isla, Year, sep = "_"))

# Matriz presencia-ausencia
beta_time_matrix <- Data %>%
  group_by(Isla_Year, Specie) %>%
  summarise(present = 1, .groups = "drop") %>%
  pivot_wider(names_from = Specie, values_from = present, values_fill = 0)

# Guardamos info de etiquetas
beta_time_labels <- beta_time_matrix$Isla_Year
beta_time_mat <- as.data.frame(beta_time_matrix[, -1])
rownames(beta_time_mat) <- beta_time_labels
```

2.4.1 — Crear matriz de presencia/ausencia por Isla y Año

```
# Matriz de distancia Jaccard entre combinaciones Isla-Año
dist_jaccard_time <- vegdist(beta_time_mat, method = "jaccard")
```

2.4.2 — Cálculo de disimilitud (Jaccard)

```
nmds_time <- metaMDS(beta_time_mat, distance = "jaccard", k = 2, trymax = 100)
```

2.4.3 — Ordenación NMDS para visualizar patrones de cambio

```
## Run 0 stress 0.0002109918
## Run 1 stress 0.0001850993
## ... New best solution
## ... Procrustes: rmse 0.04746006 max resid 0.4561123
## Run 2 stress 0.000189659
## ... Procrustes: rmse 0.02729365 max resid 0.2733217
## Run 3 stress 0.0001984415
## ... Procrustes: rmse 0.07223969 max resid 0.6267521
## Run 4 stress 0.0001783134
## ... New best solution
## ... Procrustes: rmse 0.06410456 max resid 0.6292839
## Run 5 stress 0.0001795415
## ... Procrustes: rmse 0.04017214 max resid 0.3705752
## Run 6 stress 0.0002045307
## ... Procrustes: rmse 0.05029808 max resid 0.4761724
## Run 7 stress 0.0002014073
## ... Procrustes: rmse 0.05087195 max resid 0.4991014
## Run 8 stress 0.0001611456
## ... New best solution
## ... Procrustes: rmse 0.07031728 max resid 0.735497
## Run 9 stress 0.0001951964
## ... Procrustes: rmse 0.04880077 max resid 0.479344
## Run 10 stress 0.0002031824
## ... Procrustes: rmse 0.06458168 max resid 0.5882923
## Run 11 stress 0.0001680359
## ... Procrustes: rmse 0.06300819 max resid 0.6234477
## Run 12 stress 0.0001745766
## ... Procrustes: rmse 0.05192036 max resid 0.5366792
## Run 13 stress 0.0001854959
## ... Procrustes: rmse 0.04285179 max resid 0.3698261
## Run 14 stress 0.0002044952
## ... Procrustes: rmse 0.04946352 max resid 0.535151
## Run 15 stress 0.0001983859
## ... Procrustes: rmse 0.07372197 max resid 0.7081481
## Run 16 stress 0.0002020962
## ... Procrustes: rmse 0.05929687 max resid 0.5327436
## Run 17 stress 0.0001766002
## ... Procrustes: rmse 0.07216735 max resid 0.8210971
## Run 18 stress 0.0001869673
```

```

## ... Procrustes: rmse 0.04381536 max resid 0.3812479
## Run 19 stress 0.0002119988
## ... Procrustes: rmse 0.04475197 max resid 0.3870628
## Run 20 stress 0.0001830805
## ... Procrustes: rmse 0.05530858 max resid 0.5771433
## Run 21 stress 0.0001999943
## ... Procrustes: rmse 0.04348898 max resid 0.4795781
## Run 22 stress 0.0001956458
## ... Procrustes: rmse 0.05125078 max resid 0.427711
## Run 23 stress 0.0001789406
## ... Procrustes: rmse 0.07364436 max resid 0.8657405
## Run 24 stress 0.0002021137
## ... Procrustes: rmse 0.07399057 max resid 0.7140079
## Run 25 stress 0.0002075641
## ... Procrustes: rmse 0.05655771 max resid 0.562202
## Run 26 stress 0.0001958677
## ... Procrustes: rmse 0.0212997 max resid 0.2075237
## Run 27 stress 0.00019652
## ... Procrustes: rmse 0.06651835 max resid 0.6181989
## Run 28 stress 0.0001837728
## ... Procrustes: rmse 0.04770797 max resid 0.4501735
## Run 29 stress 0.0002106381
## ... Procrustes: rmse 0.07296027 max resid 0.6815959
## Run 30 stress 0.000215628
## ... Procrustes: rmse 0.05855303 max resid 0.636052
## Run 31 stress 0.0001646569
## ... Procrustes: rmse 0.04951099 max resid 0.3992608
## Run 32 stress 0.0001980396
## ... Procrustes: rmse 0.06126486 max resid 0.711297
## Run 33 stress 0.0002131044
## ... Procrustes: rmse 0.06442707 max resid 0.550972
## Run 34 stress 0.0002012318
## ... Procrustes: rmse 0.06907122 max resid 0.7364273
## Run 35 stress 0.0001695077
## ... Procrustes: rmse 0.05809393 max resid 0.5040199
## Run 36 stress 0.0001878329
## ... Procrustes: rmse 0.07174819 max resid 0.7351945
## Run 37 stress 0.0002002706
## ... Procrustes: rmse 0.06742734 max resid 0.7525095
## Run 38 stress 0.0001976139
## ... Procrustes: rmse 0.06812923 max resid 0.6545298
## Run 39 stress 0.00019014
## ... Procrustes: rmse 0.03317965 max resid 0.3084796
## Run 40 stress 0.0001994661
## ... Procrustes: rmse 0.04525623 max resid 0.4222558
## Run 41 stress 0.000190652
## ... Procrustes: rmse 0.04913176 max resid 0.4597681
## Run 42 stress 0.0002058206
## ... Procrustes: rmse 0.06296786 max resid 0.6497509
## Run 43 stress 0.0001999231
## ... Procrustes: rmse 0.04935878 max resid 0.4669926
## Run 44 stress 0.0002073501
## ... Procrustes: rmse 0.02220178 max resid 0.1779627
## Run 45 stress 0.0002041594

```

```

## ... Procrustes: rmse 0.04019302 max resid 0.4197329
## Run 46 stress 0.0002039506
## ... Procrustes: rmse 0.07114413 max resid 0.6357874
## Run 47 stress 0.0002101463
## ... Procrustes: rmse 0.03468841 max resid 0.2818476
## Run 48 stress 0.0001948149
## ... Procrustes: rmse 0.06941081 max resid 0.6595365
## Run 49 stress 0.0001878115
## ... Procrustes: rmse 0.07321549 max resid 0.6647877
## Run 50 stress 0.000205594
## ... Procrustes: rmse 0.06364827 max resid 0.6696568
## Run 51 stress 0.0001990205
## ... Procrustes: rmse 0.02685335 max resid 0.2660151
## Run 52 stress 0.0002113728
## ... Procrustes: rmse 0.03907828 max resid 0.336322
## Run 53 stress 0.0002078915
## ... Procrustes: rmse 0.05029432 max resid 0.4656569
## Run 54 stress 0.0001728671
## ... Procrustes: rmse 0.05991917 max resid 0.5337626
## Run 55 stress 0.0002003012
## ... Procrustes: rmse 0.04322175 max resid 0.3952416
## Run 56 stress 0.0001957887
## ... Procrustes: rmse 0.02957587 max resid 0.3116438
## Run 57 stress 0.0001941456
## ... Procrustes: rmse 0.05457972 max resid 0.5558609
## Run 58 stress 0.000171966
## ... Procrustes: rmse 0.02845931 max resid 0.2471762
## Run 59 stress 0.0001851233
## ... Procrustes: rmse 0.02986289 max resid 0.3086812
## Run 60 stress 0.0001973661
## ... Procrustes: rmse 0.06110977 max resid 0.5428933
## Run 61 stress 0.0001820201
## ... Procrustes: rmse 0.06777117 max resid 0.6752645
## Run 62 stress 0.000183858
## ... Procrustes: rmse 0.06411374 max resid 0.5964483
## Run 63 stress 0.0002014985
## ... Procrustes: rmse 0.03737234 max resid 0.356634
## Run 64 stress 0.0002028863
## ... Procrustes: rmse 0.06317877 max resid 0.7101831
## Run 65 stress 0.0001754947
## ... Procrustes: rmse 0.06148705 max resid 0.7084801
## Run 66 stress 0.0001924717
## ... Procrustes: rmse 0.03050526 max resid 0.2985605
## Run 67 stress 0.0001906494
## ... Procrustes: rmse 0.06121608 max resid 0.5312067
## Run 68 stress 0.0001873491
## ... Procrustes: rmse 0.0751181 max resid 0.6948977
## Run 69 stress 0.0001958296
## ... Procrustes: rmse 0.06786726 max resid 0.6227283
## Run 70 stress 0.0001734698
## ... Procrustes: rmse 0.07530736 max resid 0.738839
## Run 71 stress 0.0001397649
## ... New best solution
## ... Procrustes: rmse 0.06663751 max resid 0.6285428

```

```

## Run 72 stress 0.000207505
## ... Procrustes: rmse 0.02992726 max resid 0.3066383
## Run 73 stress 0.0002058945
## ... Procrustes: rmse 0.05999759 max resid 0.6673278
## Run 74 stress 0.0001883796
## ... Procrustes: rmse 0.06725875 max resid 0.6767644
## Run 75 stress 0.0001988672
## ... Procrustes: rmse 0.06664979 max resid 0.7144127
## Run 76 stress 0.0002007767
## ... Procrustes: rmse 0.07240302 max resid 0.5885585
## Run 77 stress 0.000201802
## ... Procrustes: rmse 0.04564089 max resid 0.4861253
## Run 78 stress 0.0002027386
## ... Procrustes: rmse 0.0373457 max resid 0.3713101
## Run 79 stress 0.0001911358
## ... Procrustes: rmse 0.04936298 max resid 0.4212956
## Run 80 stress 0.000197474
## ... Procrustes: rmse 0.06549224 max resid 0.7303059
## Run 81 stress 0.0002007535
## ... Procrustes: rmse 0.05525337 max resid 0.5875768
## Run 82 stress 0.0001895809
## ... Procrustes: rmse 0.05495074 max resid 0.6662395
## Run 83 stress 0.000188555
## ... Procrustes: rmse 0.0474159 max resid 0.4034124
## Run 84 stress 0.0001650257
## ... Procrustes: rmse 0.07062644 max resid 0.6192965
## Run 85 stress 0.0001621542
## ... Procrustes: rmse 0.05372776 max resid 0.5640224
## Run 86 stress 0.0001985902
## ... Procrustes: rmse 0.04913791 max resid 0.3714762
## Run 87 stress 0.0001576837
## ... Procrustes: rmse 0.04867871 max resid 0.3934697
## Run 88 stress 0.0001459436
## ... Procrustes: rmse 0.05624155 max resid 0.5645861
## Run 89 stress 0.0001796517
## ... Procrustes: rmse 0.05882632 max resid 0.6058964
## Run 90 stress 0.000183963
## ... Procrustes: rmse 0.07198402 max resid 0.7414258
## Run 91 stress 0.0002004156
## ... Procrustes: rmse 0.06836939 max resid 0.7466361
## Run 92 stress 0.0002065368
## ... Procrustes: rmse 0.07057252 max resid 0.6585364
## Run 93 stress 0.0001909059
## ... Procrustes: rmse 0.0522409 max resid 0.501459
## Run 94 stress 0.0001933306
## ... Procrustes: rmse 0.05645414 max resid 0.6824736
## Run 95 stress 0.0001693161
## ... Procrustes: rmse 0.05983238 max resid 0.6832799
## Run 96 stress 0.0002017151
## ... Procrustes: rmse 0.04307909 max resid 0.5109837
## Run 97 stress 0.0001740565
## ... Procrustes: rmse 0.04781844 max resid 0.4977228
## Run 98 stress 0.0002109042
## ... Procrustes: rmse 0.07156047 max resid 0.6813904

```



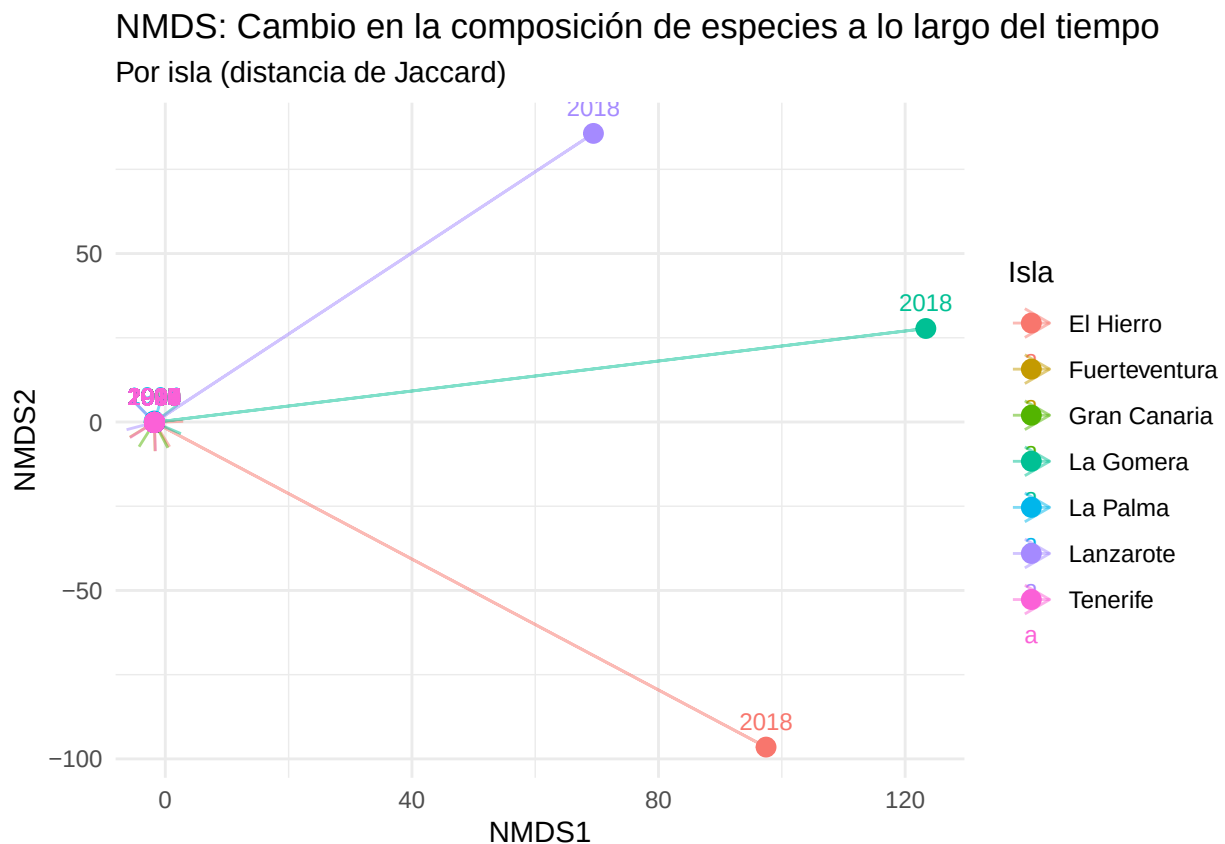
```
## Run 99 stress 0.0001667664
## ... Procrustes: rmse 0.03867305 max resid 0.4036394
## Run 100 stress 0.0001824401
## ... Procrustes: rmse 0.03460164 max resid 0.3790901
## *** Best solution was not repeated -- monoMDS stopping criteria:
## 100: scale factor of the gradient < sfgrmin
```

```
# Extraer coordenadas
```

```
nmds_coords <- as.data.frame(scores(nmds_time, display = "sites"))
nmds_coords$Isla_Year <- rownames(nmds_coords)
nmds_coords <- nmds_coords %>%
  separate(Isla_Year, into = c("Isla", "Year"), sep = "_")
```

```
# Gráfico NMDS temporal
```

```
ggplot(nmds_coords, aes(x = NMDS1, y = NMDS2, color = Isla, label = Year)) +
  geom_path(aes(group = Isla), arrow = arrow(type = "open", length = unit(0.15, "inches")), alpha = 0.5) +
  geom_point(size = 3) +
  geom_text(size = 3, vjust = -1) +
  theme_minimal() +
  labs(title = "NMDS: Cambio en la composición de especies a lo largo del tiempo",
       subtitle = "Por isla (distancia de Jaccard)", x = "NMDS1", y = "NMDS2")
```



3. Análisis de Diversidad Funcional y Taxonómica

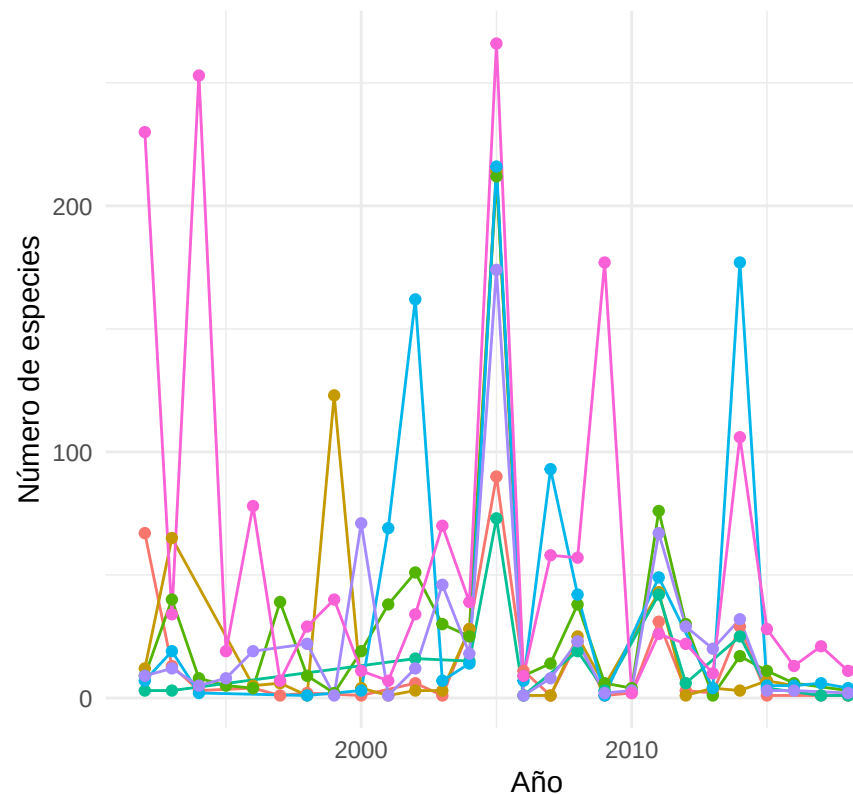
Aquí distinguiremos dos ramas:

3.1 Diversidad taxonómica: basada en recuentos y relaciones jerárquicas (Filo, Clase, Orden, etc.)

```
# Riqueza de especies por Isla y Año
richness_taxa <- Data %>%
  group_by(Isla, Year) %>%
  summarise(Riqueza = n_distinct(Specie), .groups = "drop")

ggplot(richness_taxa, aes(x = as.numeric(as.character(Year)), y = Riqueza, color = Isla)) +
  geom_line() +
  geom_point() +
  theme_minimal() +
  labs(title = "Riqueza taxonómica por isla a lo largo del tiempo",
       x = "Año", y = "Número de especies")
```

Riqueza taxonómica por isla a lo largo del tiempo



3.1.1 Preparar tabla de riqueza taxonómica

3.1.2 Diversidad taxonomica con jerarquía

```
# Matriz con jerarquía taxonómica
taxonomy_df <- Data %>%
  select(Specie, Filo, Clase, Orden, Familia, Genero) %>%
```

```

distinct() %>%
column_to_rownames("Specie")

# Distancia taxonómica
taxo_dist <- taxa2dist(taxonomy_df, varstep = TRUE)

# Aplicar taxondive a una comunidad ejemplo (por Isla y Año)
comunidad_ejemplo <- Data %>%
  filter(Isla == "Tenerife", Year == "2008") %>%
  count(Specie) %>%
  pivot_wider(names_from = Specie, values_from = n, values_fill = 0)

# Convertir a matriz
comunidad_mat <- as.matrix(comunidad_ejemplo)

# Calcular diversidad taxonómica
taxo_div <- taxondive(comunidad_mat, taxo_dist)
print(taxo_div)

```

```

##           Species   Delta  Delta* Lambda+  Delta+ S Delta+
##           57.000  66.758  84.441 593.080  81.650    4654
## Expected           69.513  66.758           81.650

```